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Journal of Immunoassay and Immunochemistry

ISSN: 1532-1819 (Print) 1532-4230 (Online) Journal homepage: https://www.tandfonline.com/loi/ljii20

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**To cite this article:** A. B. Damola, J. A. Adeniji & A. S. Bakarey (2019): Hepatitis C virus seropositivity and the risk factors for transmission among blood donors in some selected centers in Lagos State, Southwest Nigeria, Journal of Immunoassay and Immunochemistry, DOI: 10.1080/15321819.2019.1647853

**To link to this article:** https://doi.org/10.1080/15321819.2019.1647853

Published online: 04 Aug 2019.

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JOURNAL OF IMMUNOASSAY AND IMMUNOCHEMISTRY
https://doi.org/10.1080/15321819.2019.1647853

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# Hepatitis C virus seropositivity and the risk factors for transmission among blood donors in some selected centers in Lagos State, Southwest Nigeria

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**ABSTRACT**

**Introduction**: Hepatitis C virus (HCV) infects about 0.5% to 2.3% of the world population with most of the cases occurring in developing countries. It is primarily transmitted through transfusion of blood and blood products. There exists dearth of information on burden and circulation of HCV and their attendant health challenges in Nigeria. This study was therefore designed to determine the seroprevalence rate and risk of HCV transmission among blood donors in Lagos State Nigeria.

**Methodology**: Blood samples were collected between January 2002 and December 2006 from 3,002 consenting (Male = 2,922; Female = 80; Age range = 18–63; Median age = 32 years) donors in five selected public hospitals' blood donation centers between 2002 and 2006. Sera was tested for anti-HCV by ELISA technique. Demographic and other relevant information were obtained by a semi-structured questionnaire to assess risk factors for HCV transmission.

**Results**: This study found an overall rate of 3.1% for anti-HCV among the blood donors sampled. Highest rate of 6.0% for HCV was found among participants age ranged ≥50 years and lowest in the age group 40–49 years. Prevalence of HCV was higher in female (6.3%) than in male (3.0%) and was 0.21 times less risky in female compared to their male counterparts (OR = 1.29, 95%CI 0.11–1.31). By location, MSCH had the highest HCV rate (3.9%) and lowest (2.1%) in GHOA. Sharing of sharps for tattoo/tribal markings had a statistical association ($p$ = .0379) with HCV infection. However, no significant difference was found by gender (CI = 0.99–2.01; $p$ = .1002), age (CI = 0.79–1.55; $p$ = .1001) and location ($p$ = .5326).

**Conclusion**: The relatively high prevalence of HCV infection detected and the risk of transmission among blood donors in this study are of public health importance. Hence, the institution of appropriate measures to stem down the trend of HCV circulation among this population in Nigeria is therefore advocated.

**KEYWORDS**
Hepatitis C virus; seroprevalence; ELISA technique; blood donors; Nigeria

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## Introduction

Hepatitis C virus (HCV) is a common cause of post-transfusion hepatitis. It has infected about 170 million people worldwide with an estimated 32 million in Africa.[1–3] In Nigeria, epidemiological information indicates that so far the prevalence of hepatitis C virus infection is on the increase, ranging from 4.7–5% in Ilorin, to 5.3–6.6% in Enugu, to 11% in Ibadan and 20% in Benin.[3] Children and adults are all at risk of being infected especially sickle cell disease patients. Others include those who are exposed to the common risk factors like Blood transfusion, hemodialysis, recycling of syringes and needles, sexual promiscuity[1] The risk of infection is high among blood transfusion and organ recipients, intravenous drug users and hemophiliac. Infection has also been associated with social and cultural practices of using percutaneous procedures such as ear piercing, body piercing, circumcision, etc., if the equipment used is not sterilized.[4,5] Transmission of HCV predominantly occurs parenterally as a result of blood transfusion and exposure to blood derivatives, actually the disease was first recognized in recipients of blood and blood products such as factor VIII and immunoglobulins. In addition, transplanted organs and needle stick injuries have also been implicated in transmission.[6] A small risk through sexual contact and low mother to baby transfer exists. Transmission has also been reported in drug misusers and patients in dialysis and surgical units.[6,7]

HCV has been implicated as the causative agent in a large proportion of sporadic cases of Transfusion Transmission viral (TTV) hepatitis. Perhaps of even greater long-term significance to public health, chronic infection with HCV has been linked to the development of hepatocellular carcinoma.[8] The presence of antibodies to various hepatitis C viral antigens in a person's serum indicates that there has been an infection with the virus, and in most cases, portrays a chronic infection.[6] Anti-HCV antibodies develop slowly up to 1–3 months after the onset of clinical illness. The antibodies which could not be detected for up to 1 year in some patients were found in only 60% of patients with sporadic TTV hepatitis.[9] Commercial enzyme immunoassays or enzyme-linked immunosorbent assays (ELISA) for detection of HCV antibodies are available using recombinant proteins or peptides as antigens. These available assays will detect chronic infections but are less sensitive in the early stages of the infection where there may be a window phase of up to 3 months before tests become positive.[6]

Early studies of HCV prevalence were done among transfusion recipients and in people with medical conditions associated with high rates of blood transfusion, such as hemophilia, thalassemia, chronic renal failure, and cardiac surgery. The results showed that the rate of Non-A Non-B hepatitis in these groups was largely related to the acquisition of HCV infection.[10–12] However, post-transfusion HCV has become relatively rare in developed countries since blood donors with the risk of blood-borne viral infections and those found to have HCV, HIV, or HBV have been excluded from

donating blood.[12] In a study at Sacramento Medical Foundation Blood Centers, California, using volunteer repeat donors carefully screened for hepatitis risk factors, and tested for HBV infection and with a sensitive assay for anti-HCV done, the risk of HBV being transfusion transmitted was in the order of 1 in 63,000 units, while for HCV it was 1 in 125,000 units. These risks were so small that it was suggested that other means of acquiring these viruses should be sought when patients develop hepatitis following blood transfusion.[13,14]

In the developed world, effective control has been put in place to stem down the circulation of HCV especially among blood donors as a result of effective diagnosis which has become a greater challenge in developing countries like Nigeria where the infection is on the rise.[1] However, in spite of several measures put in place to control HCV transmission in advanced countries, a further challenge is continued transmission of HCV infection in high-risk population specially injecting drug users (IDUs) as the major reservoir of the infection (Farshadpour et al. 2016). This continuous rise in the number of IDUs along with the rising prevalence of HCV infection among young IDUs is an indication of a forthcoming public health problem, presenting a serious challenge to control the transmission of HCV infection.[4,9]Though post transfusion hepatitis C has substantially decreased in the developed countries, since the addition of anti-HCV antibody assay to the screening regimen of blood donors, the same cannot be said of developing countries where anti-HCV antibody screening is not mandatory and hence not routinely done. Furthermore, the WHO recommendation that blood and blood products be screened to prevent the earlier discovered pathogens such as HBV was complied with by only six African countries, while screening in 18 countries was done only in large cities, and in other countries, no screening was done.[14,15] There exists dearth of information on burden and circulation of HCV due to its asymptomatic nature and the attendant health challenges in Nigeria. This study was therefore designed to determine the seroprevalence rate of HCV among blood donors in Lagos State Nigeria.

## Methodology

### *Area of study and distribution of samples from blood donation centers in lagos state*

Three thousand and two blood donor samples, from 2922 males and 80 females, were collected from the blood donation centers of five[5] major public hospitals in Lagos State between January 2002 and December 2006. The highest number of blood samples 808 (26.9%) comprised of 796 males and 12 females were from the General Hospital Lagos; while the least number of samples, 237 (7.9%) from 229 males and 8 females were from the General Hospital Gbagada. The Lagos

State University teaching Hospital had 767 (25.5%) blood samples made up of 767 from males and 8 from females. General Hospital Orile Agege was the source of 466 (15.5%) blood samples: 456 from males and 10 from females. There were 714 blood donor samples from the Massey Street Children's Hospital, 672 from male blood donors and 42 from females.

### Sample collection, transportation and lab analysis

A semi-structured questionnaire was used to capture all demographic and all other relevant information including risk factors associated with HCV transmission to assess their knowledge about the virus before specimen collection. This semi-structured questionnaire was self-administered except for those who required assistance due to their level of education. Venous blood specimens (5 mls) were aseptically collected from each of the 3002 blood donors in the Blood donation centers in the five public hospitals in Lagos State, into dry sterile-labeled specimen bottles with the donor's laboratory identity number and date of collection. The blood was allowed to clot and then spun after the clot retracted by centrifugation at 2000rpm for 20 min. The sera was then separated from blood cells within 4 hours and stored in two aliquots (A and B) each at $-70^{O}$C until assays were done. The aliquot A was screened for HCV antibodies by ELISA technique. Prior to the specimen collection, ethical approval was obtained from Lagos State University Teaching Hospital (LASUTH) Health Research and Ethics committee (LRECO6/10/1058). The laboratory analysis was carried out in LASUTH and the Department of Virology, University College Hospital (UCH), Ibadan. A cold chain was maintained during the transportation of the samples. Reagents from Murex anti-HCV (version 4.0) kit of Abbot Diagnostic Division, manufactured by Murex Biotec S. A. (Pty) Ltd. Republic of South Africa were used for the detection of anti-HCV. The assay was performed in accordance with the manufacturer's instructions. Optical density (OD) was read using the Emax endpoint ELISA microplate reader (Molecular Devices, California, USA) and subsequently interpreted in line with the manufacturer's instruction. Participation in this study was voluntary and all persons who did not consent at any stage of the study recruitment were excluded in order to follow the principle of autonomy. They were provided with full assurance of anonymity, confidentiality, privacy together with all the information supplied including the samples to be collected.

### Data analysis

Data were analyzed using SPSS version 21. Mean, median figure and tables were used to describe the results using Chi-square, multiple logistic regression and ANOVA at 95% CI for Odds Ratio while $p < .05$ was considered significant.

## Results

Clinical correlates of HCV rates associated with sociodemographic information of the study participants showed that 991 (33.0%) participants had prior knowledge of hepatitis C virus while 2011 (67.0%) were ignorant with 952 (31.7%) claiming to have been involved in multiple sexual partnerships and 2050 (68.3%) had either one sexual partner or did not participate in sexual activity. One thousand nine hundred and ninety-two (68.4%) of them had no incisions on their bodies while 1010 (33.6) were incised at one point in their lives. There was a statistical association ($p$ = .0379) between HCV infection and sharing sharp objects or incision for tattoo (Table 1). A total of 661 (22.0%) of the participants had only primary education, 103 (3.4%) did not have any formal education while the rest 1536 (51.2%) and 712 (23.7%) had secondary and tertiary education, respectively. The participants were in the category of 874 (29.1%) individuals as singles and not married, 1554 (51.8%) were married while 561 (18.7%) were separated from their partners. However, only 13 (0.4%) of the participants were widowed. The prevalence rate of HCV was higher in female (6.3%) than in male (3.0%) and was 1.3 times higher in female compared to their male counterparts (OR = 1.29, 95%CI 0.99–2.01). Of the 2922 male participants tested 87 (3.0%) were positive for HCV infection while 5 (6.3%) female participants tested were positive. However, there was no statistical association by gender with respect to HCV infection in this study ($p$ = .1002) (Table 1). Furthermore, distribution of HCV infection by marital status in the study community shows that highest prevalence (3.7%) was detected among married participants followed by the single (2.5%) and the lowest rate (2.1%) was found among the separated (Table 1). An overall rate of 3.1% of HCV infection was reported in this study. Highest HCV rate (6.0%) was detected in the age group >50 years while the lowest rate (1.9%) of HCV infection was found in the age group 40–49 years (Figure 1). However, none of the participants in the widowed participants was positive for HCV infection in this study (table 1). Participants were drawn from five blood donation centers for this study. Highest HCV infection rate (3.9%) was found among participants in MSCH followed by LASUTH while the lowest rate of 2.4% was in GHOA (Table 2).

## Discussion

This study found an overall HCV rate of 3.1% among blood donors in Lagos state, Nigeria. The rate falls within 0.5% to 8% but lower than 3.9%–12.8% range reported in some studies carried out among adult population in Nigeria[16] (Gower *et al.*, 2014) and in consonance with 3% prevalence reportedly found among blood donors worldwide.[3,17,18] It is significantly higher than the prevalence of <1% HCV found in Western Europe and North America[17], where post-transfusion HCV has become relatively rare, since blood donors with the risk of blood-borne viral infections and those found to have HCV, HIV, or HBV

**Table 1. Sociodemographic information and clinical correlates associated with HCV infection among the study population in Lagos State**: Results of univariate and multivariate logistic regression analyses are presented below.

| Characteristics | No Tested (N=3002) | No (%) Positive | p-value | OR (95% CI) |
|---|---|---|---|---|
| **Age range (years)** | | | | |
| <20 | 32(1.1) | 1(3.3) | 0.1001 | |
| 20-29 | 770(25.6) | 20(2.6) | | 2.39 (0.91-6.09) |
| 30-39 | 1615(53.7) | 56(3.5) | | 3.29 (1.40-7.22) |
| 40-49 | 521(17.4) | 14(2.7) | | 1.76 (0.60-5.11) |
| ≥50 | 64(2.1) | 2(3.1) | | 1.11 (0.35-3.51) |
| **Marital Status** | | | | |
| Single | 874(29.1) | 22(2.5) | 0.718 | |
| Married | 1554(51.8) | 58(3.7) | | 2.42 (1.10-5.29) |
| Separated | 561(18.7) | 12(2.1) | | 1.60 (0.65-3.66) |
| Widowed | 13(0.4) | 0(0.0) | | 0.48 (0.19-1.26) |
| **Gender** | | | | |
| Male | 2922(97.3) | 87(3.0) | 0.1002 | 1.29(0.99-2.01) |
| Female | 80(2.7) | 5(6.3) | | |
| **Level of Education** | | | | |
| Primary | 661(22.0) | 13(2.0) | 0.922 | |
| Secondary | 1536(51.2) | 45(2.9) | | 3.35 (1.38-8.38) |
| Tertiary | 712(23.7) | 22(3.1) | | 0.72 (0.16-3.27) |
| None | 103(3.4) | 10(9.7) | | 2.13 (1.11-4.08) |
| **Knowledge about hepatitis C infection** | | | | |
| Yes | 991(33.0) | 14(1.4) | 0.0581 | |
| No | 2011(67.0) | 78(3.9) | | |
| **Having multiple sexual partners** | | | | |
| Yes | 952(31.7) | 41(4.3) | 0.0735 | |
| No | 2050(68.3) | 51(2.5) | | |
| **Have tattoo, incision/tribal in any part of the body** | | | | |
| Yes | 1992(66.4) | 24(1.2) | **0.0379*** | 1.34(1.01-2.23) |
| No | 1010(33.6) | 78(7.7) | | |

infections had been excluded from donating blood.[19,20] This study reported that a greater proportion of the participants (67.0%) did not have any knowledge about HCV of which 3.9% of them were positive. However, this risk factor was not statistically significant (0.0581) and in concordance with the previous report.[3] A higher level of public enlightenment about the HCV infection has reduced the activity of scourge to a low level in the developed world when compared with developing countries in Africa and Southeast Asia.[21–24] However, considerably higher HCV prevalence ranging from 10% to 20% were earlier found among patients in Nigeria and other African countries due to ignorance on the mode of its transmission.[3,25,26]

Stratified by age groups, the number of blood donors declined with increasing age but conversely with increased prevalence of HCV infection. Thus, the oldest age group ≥50 years had the highest prevalence of 6.0%. This is in contrast to the report of Jeremiah *et al.*[27] in which age group 21–30 years had the highest HCV prevalence of 60% in Port Harcourt. Though the relatively fewer number of donors

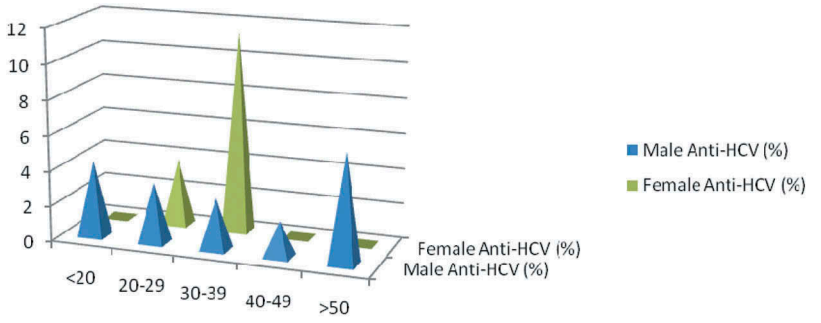



**Figure 1.** Age and gender distribution of Anti-HCV among blood donors in Lagos State.

**Table 2.** Overall distribution of HCV infection by location in Lagos State.

| LOCATION | NO TESTED | NO (%) POSITIVE |
|---|---|---|
| GHL | 808 | 22(2.7) |
| LASUTH | 777 | 25(3.3) |
| MSCH | 714 | 28(3.9) |
| GHOA | 466 | 11(2.4) |
| GHG | 237 | 6(2.5) |
| TOTAL | 3002 | 92(3.1) |

$\chi 2_{0.05:1} = 3.15$; $p = 0.5326$

tested in this age group might be a factor for the higher prevalence. The 6.0% prevalence nevertheless indicates that HCV has been active in this age group and this is in agreement with an earlier study by Forbi *et al.*[26] which reported that the virus had not been recently introduced into the Nigerian environment. The prevalence declined to a lower rate of 1.9% among 40–49 years age group. This reduction in HCV prevalence to this stage can be seen to have divided the blood donors into two broad groups. The two upper age groups with sharply declining HCV rates, and the lower three age groups in which the prevalence rates are slightly increasing. Various risk factors for HCV transmission identified in this study included sharing of sharps for tattooing/scarification (OR = 0.64 95%CI 0.15–1.47; $p = .0379$) found to significantly associated with HCV.[28,29] Thus, there seem to be greater risks of HCV transmission with blood from the younger donors (the vast majority) who constituted 83.7% of the blood donors. This agrees with a previous study in Port Harcourt in which donors in the age bracket 21–40 constituted 80% of the positive donors.[27]

The difference in HCV antibodies prevalence between the male (3.0%) and female (6.2%) donors, which was not statistically significant ($\chi 2_{0.05:1} = 1.47$; $p = .1001$) may be probably due to the evident male bias in blood donation. Despite the higher HCV rate in female than male, it is 0.21 times less risk of contracting HCV infection in female compared to their male counterparts (OR = 0.21, 95%CI 0.11–1.31). Also, due to this bias, the percentage of female donors (2.7%) tested

in this study was relatively much lower than the (97.3%) male donors tested for anti-HCV. This disparity also found in other studies on HCV prevalence among blood donors reported by Ayolabi *et al.*[30] in Lagos had 97% male and 3% female blood donors, Jeremiah *et al.* (2008) in Port Harcourt reported 88% males and 12% females, Bala *et al.*[23] in Kano had 99.7% male and 0.3% female donors tested and Martins *et al.*[31] in Brazil reported 95% males to 5% females tested. This could be a factor in the relatively high prevalence obtained for the female donors. According to Bala *et al.*[23], the much higher number of male donors compared to their female counterparts could be as a result of the inability of women to donate blood because of some physiological factors such as menstruation and pregnancy which limit the chances of potential female donors. Hence, the high disparity in the number of male and female donors in this study could be a factor in the higher HCV prevalence obtained for the female donors when compared with male donors supported by earlier reported.[14]

The differences in HCV prevalence in the five locations used in this study were not statistically significant ($p$ = .5326). This conformed to the work of Kawo *et al.*[32] who found no significant difference in HCV prevalence among blood donors in five locations in Kano. On the contrary, there was a significant difference ($p$ = .05) in a similar study in Brazil by Martins *et al.*[31] That study however, covered a much larger territory which constituted three regions of Brazil. The slight differences observed in this study may be due to the fact that the locations are under the jurisdiction of the same Health authority that applies uniform processes, and procedures in the selection of blood donors. In a well-organized system, where there are guiding regulations for HCV testing of blood donors and other individuals, those with HCV antibodies can be detected in any location within the territory.[3] However, in the situation that obtains in Nigeria where anti-HCV testing is generally not performed in many transfusion centers[33] unknown and uninformed anti-HCV positive donors, and those not properly counseled may evade detection in other localities. The blood units donated by such donors, if transfused will result in HCV being transmitted to the recipients. A mandatory national law and national algorithm designed for HCV testing in the country will ensure that blood donors are tested uniformly across all localities in the country as it is practiced in the developed world.[3,24,31]

## Conclusion

The donors were apparently free of symptoms and were therefore deemed qualified to donate blood. This portends great risks to the general populace, as the transmission of HCV predominantly occurs parenterally as a result of blood transfusion and exposure to blood derivatives. Patients transfused with blood from the seropositive donors, some of whom may also be viraemic, are subject to direct transmission of the hepatitis C virus. This study also identified the sharing of sharps for tattooing and scarification as statistically

associated with HCV transmission among others. The fact that a donor may donate blood, 6 months after a previous donation, further puts more transfusion recipients at risk of direct transmission with the hepatitis C virus. Risks to health-care workers such as blood donation officers (with other transmissible viral infections), who bleed such donors are also much increased.

## Strength and limitations of the study

Many females in comparison to their male counterparts did not turn up and only 80 out of the females consented to participate in the study and were subsequently recruited into the study. Also, power estimation was not done as regards the number of female and male participants to be recruited for the study knowing that there are few female donors available in our blood donation centers in Lagos State, Nigeria. Our strength was in the area of providing education and awareness on the mode of HCV transmission by assessing the risk factors associated with it. The next stage of this study is the detection of HCV RNA and the molecular characterization which is ongoing.

## Acknowledgments

The authors wish to thank all the participants for their cooperation. They express their gratitude to the management, directors, and staff of the five blood donation centers in Lagos including the Lagos State Health Management Board for their approval for the study.

## Declaration of conflict of interest

Authors declare that there is no conflict of interest.

## Authors' contributions

(a) Study Design (All authors)
(b) Sample collection (DAB, BAS, AJA)
(c) Reagent acquisition and Laboratory and Data analysis (All authors)
(d) Wrote the first draft of the manuscript (DAB)
(e) Revised the manuscript (All authors)
(f) Read and approved the final draft (All authors)

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